

Lab 4 Understanding Labels, Selectors and Annotations in Kubernetes

Objective

The objective of this lab is to understand:

- What are Labels in Kubernetes
 - What are Selectors in Kubernetes
 - What are Annotations in Kubernetes
 - Difference between Labels and Annotations
 - How Selectors use Labels
 - How Kubernetes identifies Pods using Labels
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What are Labels?

Labels are key-value pairs attached to Kubernetes resources.

They are used to:

- Identify resources
- Organize resources
- Group resources together
- Select resources using Selectors

Example:

```
labels:  
  app: nginx  
  env: dev
```

In this example:

```
app = nginx  
env = dev
```

The Pod can be identified as:

```
Application = nginx  
Environment = dev
```

Labels act like tags attached to a resource.

What are Selectors?

Selectors are used to select Kubernetes resources based on Labels.

Example:

```
selector:  
  app: nginx
```

Kubernetes will look for:

All Pods having:

```
app = nginx
```

Only matching Pods will be selected.

Labels and Selectors Relationship

Consider three Pods:

```
Pod-1  
app=nginx
```

```
Pod-2  
app=nginx
```

```
Pod-3  
app=apache
```

Selector:

```
selector:  
  app: nginx
```

Result:

```
Selected:  
Pod-1  
Pod-2
```

```
Not Selected:  
Pod-3
```

This is the same mechanism used by:

- Services
- ReplicaSets
- Deployments

to find their Pods.

What are Annotations?

Annotations are also key-value pairs attached to Kubernetes resources.

Unlike Labels, they are not used for selection.

They are used to store additional information.

Example:

```
annotations:  
  owner: vikash  
  purpose: learning
```

Annotations are metadata.

Selectors cannot use Annotations.

Labels vs Annotations

Labels	Annotations
Used for identification	Used for additional information
Used by Selectors	Not used by Selectors
Used for grouping resources	Used for metadata storage
Small identifying information	Detailed descriptive information

Create a Pod with Labels and Annotations

Create a file:

```
pod-labels.yaml
```

Add the following configuration:

```
apiVersion: v1
kind: Pod

metadata:
  name: nginx-pod

  labels:
    app: nginx
    env: dev

  annotations:
    owner: vikash
    purpose: learning

spec:
  containers:
  - name: nginx-container
    image: nginx
```

Create the Pod

Command:

```
kubectl apply -f pod-labels.yaml
```

Expected Output:

```
pod/nginx-pod created
```

Verify the Pod

Command:

```
kubectl get pods
```

Expected Output:

NAME	READY	STATUS
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```
nginx-pod    1/1    Running
```

View Labels

Command:

```
kubectl get pods --show-labels
```

Expected Output:

NAME	READY	STATUS	LABELS
nginx-pod	1/1	Running	app=nginx, env=dev

Observe:

```
app=nginx  
env=dev
```

These are Labels attached to the Pod.

Select Pod Using Label Selector

Command:

```
kubectl get pods -l app=nginx
```

Expected Output:

NAME	READY	STATUS
nginx-pod	1/1	Running

Explanation:

-l means Label Selector

app=nginx means:

Show Pods having
app=nginx

View Annotations

Command:

```
kubectl describe pod nginx-pod
```

Look for:

```
Annotations:  
  owner: vikash  
  purpose: learning
```

These are annotations stored with the Pod.

Delete the Pod

Command:

```
kubectl delete pod nginx-pod
```

Expected Output:

```
pod "nginx-pod" deleted
```

Key Learning Points

- Labels are used for identification.
 - Labels are key-value pairs.
 - Selectors use Labels to select resources.
 - Annotations store additional metadata.
 - Annotations cannot be used by Selectors.
 - Services use Selectors to identify Pods.
 - Deployments use Selectors to manage Pods.
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Interview Questions

Q1. What are Labels in Kubernetes?

Labels are key-value pairs attached to Kubernetes resources for identification and grouping.

Q2. What are Selectors?

Selectors are used to select Kubernetes resources based on matching Labels.

Q3. What is the difference between Labels and Annotations?

Labels are used for identification and selection.

Annotations are used to store additional metadata.

Q4. Can Selectors use Annotations?

No.

Selectors can only work with Labels.

Q5. Why are Labels important?

Labels help Services, ReplicaSets and Deployments identify and manage the correct Pods.